Mapping the Green: A Spatial Survey of Campus Greenspaces at Texas A&M

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Abstract

We are planning on looking around campus and finding the variously sized greenspaces/gardens on campus in order to document them. This could be used for documentation of environmental risks, mental health, and aesthetics. There can be many risks when it comes to spaces that are at a flood risk. One risk is that it can cause blockages which lead to roads being shut down, buildings being cut off, and etc. There are also mental health issues that can be alleviated or otherwise lessened by the presence of green spaces on campus. Currently the lack of information regarding greenspaces leaves us unable to address these issues accurately. Our goal is to close that gap in information to allow changes to be made on campus where necessary to address these issues.

Introduction

PROBLEM: Texas A&M University's greenspaces lack accurate documentation

- Why this matters:

- **Greenspaces help to mitigate flood risk by storing and slowing water.**
- **They promote student mental health, stress management, and campus aesthetics.**
- **Without documentation, the university cannot determine whether we have enough green space or which areas require improvement.**

Goal: Develop a comprehensive GIS-based inventory of green spaces on the TAMU main campus.

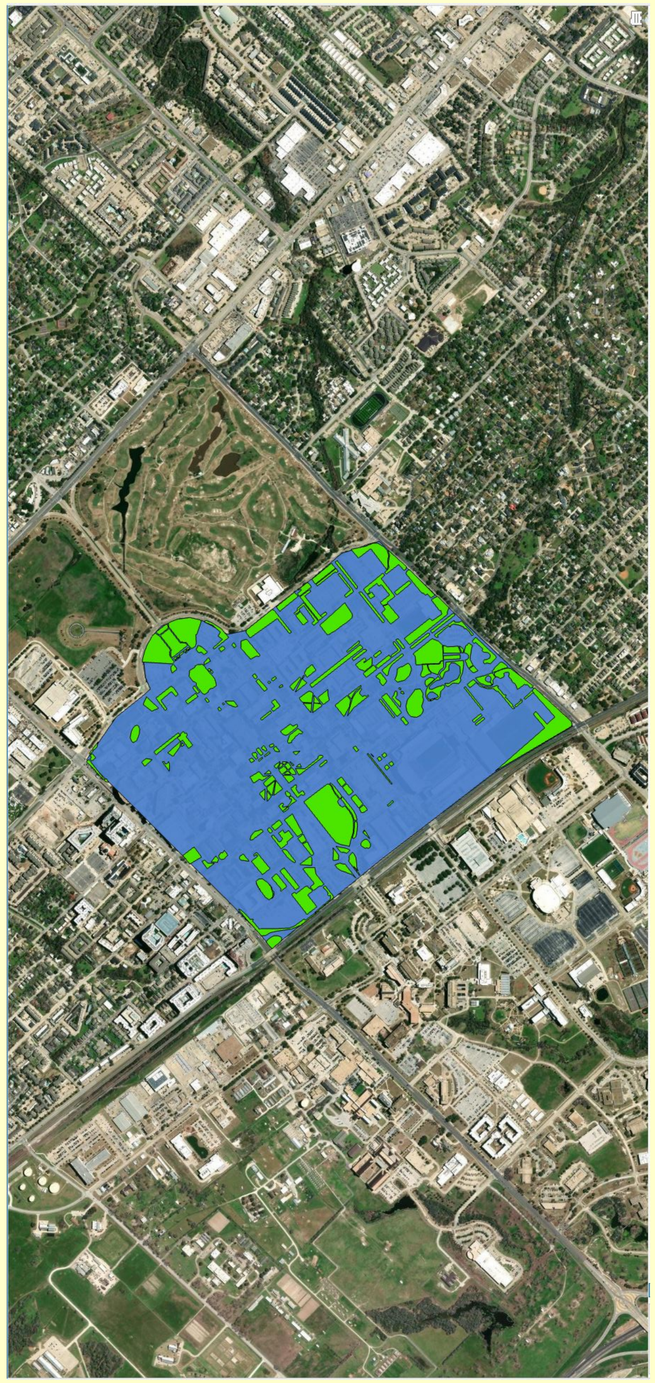
Methods

What were the criteria a space would need to meet in order for us to call it a green space? The space needed:

- **To be at least 10x10 ft (~3x3 m)**
- **Some form of permeable material like dirt, sand or gravel**
- **To not be located on or inside of any building**
- **At least one type of real vegetation**
- **The ArcGIS Field Maps app was used to collect data for this feature layer**
- **Each team member went to different parts of the study area in order to collect data**

Because we used ArcGIS field maps we were all able to contribute to the project, whether we were together or separate. This allowed us to easily split up the work and make data collection much more manageable.

Results



Discussion

Studies consistently find that within a city, Green spaces reduce runoff volumes and peak flows during storms, because trees, soil and vegetation absorb and slow water. A global study of urban green space found that larger amounts of dispersed green patches significantly mitigated flooding risk compared to the same area with more roofs and paving.

Based on our data collection, roughly 20% of Texas A&M's main campus area is green space suitable to provide a noticeable effect on rainfall runoff. This is on the low end when it comes to metropolitan areas, with places like New York having a higher percentage of green space, leaving us to conclude that A&M's campus could do with more green spaces to help mitigate disturbances and flood risk provided by rainfall.

But it's not just about total percentage of green space available, it's also about distribution. There are areas around the main Texas A&M campus that have little to no permeable green space, such as around Blocker and ZACH as well as near Kyle Field, leading to significantly higher flood risk in these areas. The administration should look into the idea of adding in more green spaces in and around these areas to help mitigate flood risk in inclimate weather.

- Green spaces have a positive relationship with mental health as individuals have less mental distress, less anxiety, and depression. There have been discussion on the greater wellbeing and healthier cortisol profiles being prevalent in urban areas with more green space compared to less.
- Green spaces are often used in a way to deliver therapeutic interventions for vulnerable groups (wilderness therapy, therapeutic horticulture, facilitated environmental conservation, care farming, ecotherapy, and etc.)
- The green spaces, as seen in the previous slide, in TAMU main campus have been areas students frequent to relax, study, and hangout to destress

Conclusion

Overall, documenting the green spaces on the Texas A&M campus provides a clear picture of how these areas benefit the campus environmentally and socially. By mapping them, we can determine where green spaces are abundant and where more are required, particularly in flood-prone areas. This project helps close an information gap and provides data that can be used to improve student well-being, reduce environmental risks, and create a more balanced and sustainable campus environment.

References

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APA Style

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